

Zachariah E. Button

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PROFESSIONAL SUMMARY

Applied ML researcher specializing in computer vision, image compression, and sensor fusion. Ten years' experience in technical R&D and data science, with a strong background in deep learning, signal processing, and traditional ML methods. Proficient in Python and PyTorch.

EDUCATION

University of Missouri - Kansas City

Jan 2022 - Present

- **Degree:** Ph.D. in Computer Science
- **Research Areas:** Efficient neural image compression, multi-sensor fusion
- **Expected:** May 2027

Kansas State University

May 2015

- **Degree:** M.S. in Statistics
- **Thesis:** Bayesian approach to apply and interpret the two-parameter Item Response Model in the context of replicated preference testing

Kansas State University

May 2013

- **Degree:** Dual B.S. in Mathematics and Statistics

COURSEWORK

Computer Science

- Computer Vision
- Deep Learning
- Multimedia Communication
- Design & Analysis of Algorithms
- Knowledge Discovery & Management
- C Programming for Engineering (C++)
- Fundamentals of Programming (Java)

Mathematics

- Optimization for Machine Learning
- Analytical Geometry & Calculus I - III
- Applied Matrix Theory (Linear Algebra)
- Advanced Calculus (Real Analysis)
- Foundations of Analysis (Real Analysis)
- Differential Equations
- Algebraic Systems (Number Theory)
- Topology

Statistics

- Statistical Theory I - II
- Statistical Computing (R, SAS)
- Experimental Design for Product Development & Improvement
- Statistical Consulting
- Numerical Methods in Statistics
- Multivariate Statistical Methods
- Theory of Life Data Analysis (Survival Analysis)
- Linear Models
- Applied Linear Statistical Models
- Categorical Data Analysis
- Biostatistics
- Analysis of Variance
- Regression & Correlation Analysis
- Probability & Statistics I - II
- Statistical Methods for Natural Sciences
- Business & Economics Statistics I - II

TECHNICAL SUMMARY

Tools

- **Python** (PyTorch, OpenCV, PySpark, Pandas, scikit-learn, Keras, XGBoost, statsmodels)
- **Deep Learning** (CNN, AE/VAE, RNN, LSTM, R-CNN, Transformers)
- **Databases** (PostgreSQL, Redshift, S3, SQL Server, SQLite, Parquet, HDF5)
- **Compute** (SageMaker, EMR, Airflow, Slurm)
- **Other** (VS Code, GitHub, Jupyter, HuggingFace, TensorBoard, W&B, Docker, R, Matlab, SAS)

Skills

- **Vision** (Classification, Detection, Tracking, Translation, Super-Resolution)
- **Deep Learning** (Optimization, Loss Function Design, Hyperparameter Tuning)
- **Traditional ML** (Regression, Forecasting, Classification, Clustering, Anomaly Detection, Survival Analysis, Dimensionality Reduction)
- **Computation** (Signal Processing, Compression, Entropy Modeling)

EXPERIENCE

Research Engineer — Jumio

May 2025 - Feb 2026

- Conducted applied research and development for upcoming liveness detection features through an exploration of computer vision and signal processing techniques

Principal Data Scientist — HP, Inc.

Feb 2022 - Mar 2025

- Led the production lifecycle of machine learning features for an enterprise fleet analytics platform serving **3M** customers, driving a sustained **30%** YoY revenue growth
- Served as the technical leader of a start-up R&D team by setting technical direction, managing cross-functional teams, mentoring junior data scientists, and cultivating a standard of collaboration
- Designed and implemented a scalable risk remediation system based on anomalous device behavior
 - Anomaly Detection stage identifies 3 types of unusual behavior across **20+** device performance metrics and **40M+** consumer devices (PySpark)
 - Risk Modeling stage maps recent anomaly patterns to risk of 4 failure events, using Cox Proportional Hazards models (scikit-survival)
 - Risk Attribution stage combines feature importance metrics with other telemetry data to identify the key factors in increased risk at the device-level
 - Remediation stage maps the key risk factors to suggested actions for reducing failure risk
- Engineered a suite of interpretable models to jointly predict device performance based on hardware and software specifications
 - Foundational models support multiple products, including simulation (effects of configuration changes on device performance) and optimization (recommendations to maximize predicted performance)
 - Two software optimization use cases deployed within 12 months for **80,000** HP devices
 - Tech stack includes SageMaker, EMR, S3, Airflow, GitHub, PySpark, Pandas, and Statsmodels
- Spearheaded the development of an “accelerators” repository, providing utility functions and data processing capabilities to improve the efficiency and accuracy of HP’s AI Systems team

Data Scientist — C2FO

Sep 2019 - Feb 2022

- Evaluated multiple optimization methods for the platform's core algorithm, demonstrating potential to increase earned revenue by **\$350k/yr** and reduce compute costs by **99%**
- Redesigned customer targeting strategy using a first principles approach and gained support from business leadership
 - Achieved **26%** lift in AUC and **70%** improvement in reliability compared to legacy system
 - Reduced running costs by **75%** and codebase by **80%**
- Developed revenue forecasting system to explain the primary drivers of C2FO market volume, achieving **3x** improvement in monthly budget accuracy

Machine Learning Engineer — Imperial PFS

Nov 2016 - Aug 2019

- Architected and built vision-based document processing system to enable **\$1M+** annual profit increase
 - Methodology included clustering, classification, object detection, OCR, NLP, and expert systems
- Executed controlled A/B testing to evaluate ML-driven customer targeting strategies, resulting in savings of over **\$900k** annually
- Automated a large variety of analytics tasks using Python, SQL, and Excel VBA, eliminating **6 hrs/week** of my own work and saving each member of the sales team over **8 hrs/week** in prospecting

Business Intelligence Developer — Epic Systems Corp.

Aug 2015 - Oct 2016

- Investigated effects of the Affordable Care Act on more than 50 utilization measures using ARIMA forecasting models and intervention analysis
- Contributed to award-winning executive-level utilization dashboard by simulating payment-level claims dataset - demonstrated at HIMSS 2016, Epic UGM 2016, and multiple customer sites

Statistical Consultant — KSU Dept. of Statistics

Aug 2014 - May 2015

- Advised university researchers on a wide variety of statistical topics, including experimental design, repeated measures analysis, logistic regression analysis, and related interpretations

RESEARCH

University of Missouri - Kansas City

May 2025 - Present

- Researcher - Multimedia Computing and Communications Lab
 - Z. Button, P. Maharjan, A.M. Badshah, and Z. Li, "SAR-PACT: Tokenization Energy-Aware Complexity Allocation for Practical Learned SAR Image Compression," (*Invited Talk*) *60th Annual Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, 2026.
 - Z. Button, P. Maharjan, and Z. Li, "Accelerating Learned SAR Image Compression via Selective Channel-Group Encoding Bypass," *IEEE International Conference on Image Processing (ICIP)*, Tampere, Finland, 2026.
 - S. Maharjan, R. S. Puttagunta, Z. Button, and Z. Li, "Cross-Modal Thermal Image Compression via RGB Side Information," *IEEE 27th International Workshop on Multimedia Signal Processing (MMSP)*, Beijing, China, 2025.

University of Pittsburgh - Summer Institute for Training in Biostatistics

2013

- Statistical analysis of adverse cardiac events in patients with Type 2 diabetes or coronary artery disease

Mount Holyoke College - Research Experience for Undergraduates

2012

- Cluster analysis for tree health in collaboration with the Harvard Forest

TEACHING

Kansas State University — M.S. Data Analytics Program

- Workshop Series Presenter
 - Machine Learning - Classification ([GitHub](#)) Mar 2022
 - Data Wrangling ([Data.World](#)) Oct 2021
 - Careers in Data Science ([Slides](#)) Sep 2020, Sep 2021
 - Machine Learning - Forecasting ([GitHub](#)) Oct 2020

Kansas State University — Machine Learning and Autonomous Systems Program

- [Instructor](#) Feb 2022 - Apr 2022
- [Advisory Board Member](#) April 2020 - Feb 2022

Kansas State University — Dept. of Statistics

- Teaching Assistant - Bus. & Econ. Statistics Aug 2013 - May 2015